Scenario-Based SQL Questions – Set 1

Practical SQL scenarios • Questions and logic explanations

Scenario 1: Employee Salary Analysis

Question:
Find the names and salaries of employees who earn more than the average salary in the company.

Logic Explanation:

First, calculate the average salary using the AVG() function.

Use a WHERE clause to filter employees whose salary is greater than the calculated average salary.

Use SELECT to display the employee name and salary.

Ensure the query uses the correct table and columns.

Concepts Used:

Aggregate Functions (AVG())

Filtering (WHERE)

Scenario 2: Customer Orders without Matching Records

Question:
Retrieve a list of customer names who have not placed any orders.

Logic Explanation:

Perform a LEFT JOIN between the customers table and the orders table using customer_id as a common field.

After the join, filter records using WHERE orders.customer_id IS NULL to find customers without any matching orders.

Select the customer_name from the customers table.

Concepts Used:

Joins (LEFT JOIN)

Filtering with NULL Values

Scenario 3: Product Sales Summary

Question:
Display the total sales amount for each product.

Logic Explanation:

Perform an INNER JOIN between the products table and the sales table using product_id.

Use the SUM() function to calculate the total sales amount for each product.

Apply a GROUP BY clause on product_name to aggregate the sales data.

Display product_name and the total sales amount.

Concepts Used:

Joins (INNER JOIN)

Aggregation (SUM())

Grouping (GROUP BY)

Scenario 4: Department-Wise Employee Count

Question:
List each department name with the number of employees working in it.

Logic Explanation:

Perform a LEFT JOIN between the employees table and the departments table using the department_id.

Use the COUNT() function to count the number of employees in each department.

Apply a GROUP BY on the department_name to aggregate the data.

Select the department name and the employee count.

Concepts Used:

Joins (LEFT JOIN)

Aggregation (COUNT())

Grouping (GROUP BY)

Scenario 5: Top 3 Highest Sales

Question:
Find the top 3 highest sales transactions.

Logic Explanation:

Select the necessary columns like sale_id, customer_id, and amount from the sales table.

Use the ORDER BY clause on the amount column in descending order to list the highest sales first.

Use LIMIT 3 to restrict the output to the top 3 records.

Concepts Used:

Sorting (ORDER BY)

Limiting Results (LIMIT)

Scenario 6: Calculate Employee Salary Ranks by Department

You have a table employees with the following columns:

employee_id (Primary Key)

name

age

salary

department_id

You also have a table departments with the following columns:

department_id (Primary Key)

department_name

Task:
Write a query to display each employee's name, department name, salary, and their salary rank within their respective department.

Logic:

Use the RANK() window function to rank employees based on their salary within each department.

Perform an INNER JOIN to combine employee data with department information using department_id.

Display employee details with their salary rank using the ORDER BY clause.